Rui Zhou

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SUMMARY

Computational materials PhD candidate (expected May 2027) with 4+ years of academic research and 2+ years of industry R&D spanning multiscale modeling, machine learning, scientific software, and automation. Combines atomistic simulation, multiscale modeling, and machine learning to investigate how structure and interfaces govern transport and performance in energy-storage and thin-film materials. Author of 13 peer-reviewed publications, developer of 4 released software applications, and inventor on 4 granted patents.

EXPERIENCE

PhD Researcher at Iowa State University *Ames, Iowa | Jan 2023 - Present*

- Perform DFT and ML-potential simulations of glassy and crystalline solid electrolytes, including Li/Na thiophosphates, closo-borates, and metal oxyhalides, to study interfaces, phase transitions, and ion transport.
- Train ML potentials on reference data from multiple electronic-structure methods, including constrained DFT for excited-state semiconductor modeling and MP2 for combustion chemistry.
- Apply Bayesian optimization to solid-electrolyte composition searches and couple an ML potential with charge equilibration (QEq) for interface studies.
- Build scientific data-analysis and machine-learning workflows with Python, NumPy, PyTorch, and MLX for large atomistic-simulation datasets.
- Build Python/Slurm workflows for HPC job preparation, execution, monitoring, and post-processing; develop agentic orchestration for atomistic simulation campaigns.
- Contribute to computational studies of ceramics, metals, and ZnS/CdTe photoplasticity using excited-state semiconductor models.
- Developed and released NAIIVE, a Swift/Metal application for 2D and spatial 3D visualization of atomistic simulations and MLIP datasets across macOS, visionOS, and iOS.

Advanced Data Storage Material Discovery Intern at Seagate Technology *Normandale, Minnesota | May 2026 - Aug 2026*

- Developed an end-to-end machine-learning workflow for thin-film ellipsometry data analysis and materials characterization.
- Built graph-neural-network and descriptor models for chemical-property prediction from learned and engineered representations.
- Applied molecular dynamics simulations to analyze atomic-scale structure and behavior in coating materials.

Simulation Engineer at Sunwoda Electric Vehicle Battery Co., Ltd. *Shenzhen, China | Dec 2020 - Dec 2022*

- Applied physics-based multiscale modeling across atomic, particle, and cell scales to support lithium-ion battery materials R&D and electrode manufacturing.
- Executed high-throughput DFT and force-field dopant screening for NCM- and LFMP-based cathodes, translating atomistic predictions into materials-development decisions.
- Developed DEM models of cathode calendaring and optimized particle-size distributions and mixing ratios to guide electrode compaction and manufacturing.
- Built autonomous parameter-identification workflows spanning DEM, P2D/ECM electrochemistry, and FEM, accelerating model calibration and simulation feedback cycles.
- Developed scikit-learn and symbolic-regression models for battery power mapping and swelling-force prediction.
- Partnered with experimental materials scientists and cell engineers to validate computational predictions, design targeted experiments, and translate material-level insights into engineering decisions.

EDUCATION

Iowa State University *Ames, Iowa | Jan 2023 - Expected May 2027*
PhD in Materials Science and Engineering

Washington University in St. Louis *St. Louis, Missouri | Aug 2018 - Jan 2020*
Master's in Materials Science and Engineering

Changchun University of Science and Technology *Changchun, China | Sep 2014 - Jun 2018*
Bachelor's in Materials Science and Engineering

SKILLS

Methods And Materials: DFT, Solid-State Materials, Molecular Dynamics, ML Interatomic Potentials, Active Learning, QEq, Graph Neural Networks, Interfaces, Ion Transport, Thin Films/Coatings, High-Throughput Screening

Selected Tools: Python, PyTorch, Swift/Metal, VASP, Quantum ESPRESSO, GAMESS, LAMMPS, ASE, Linux/Slurm/HPC